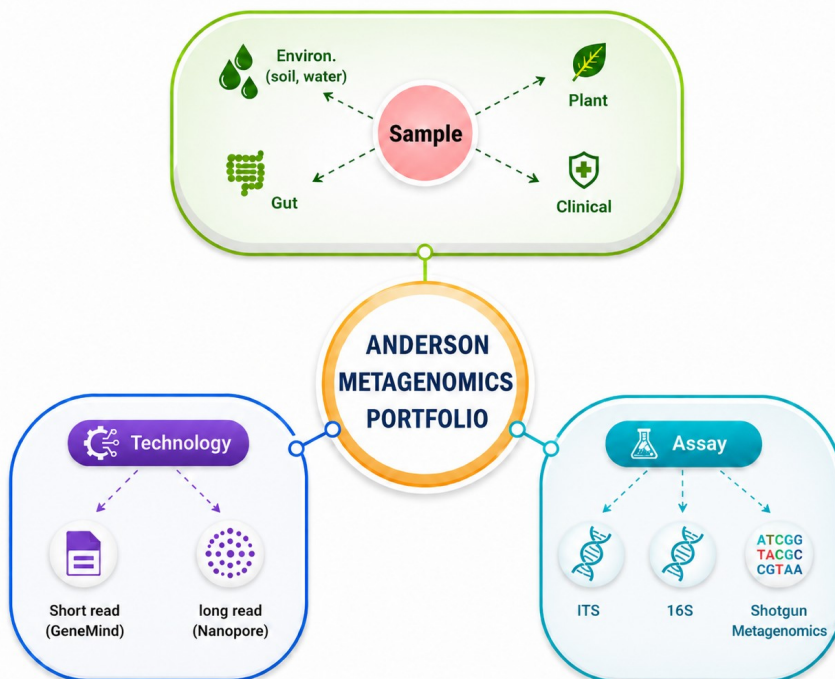


METAGENOMICS SEQUENCING SERVICES

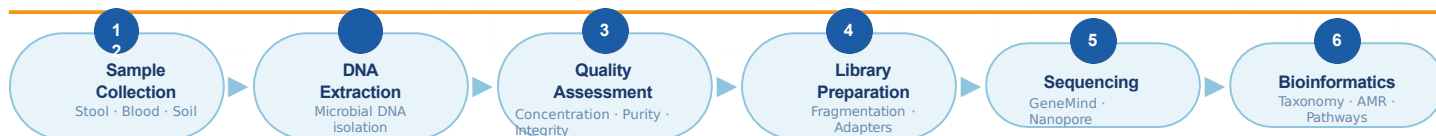
- Sequences the complete microbial community directly from clinical, agricultural, and environmental samples — no culture required.
- Simultaneous identification of bacteria, fungi, viruses, archaea, and protozoa from single sequencing run.
- Integrated AMR gene profiling, functional pathway analysis, and strain-level resolution in one validated workflow.

Anderson Metagenomics Panel — Whole Metagenome Shotgun | 16S rRNA Community Profiling | ITS Mycobiome | Clinical Pathogen Detection (blood, CSF, BAL, stool) | Gut & Urobiome Analysis | Plant & Rhizosphere Profiling | Environmental Surveillance | Nanopore Long-Read Assembly | AMR Genotyping.



Sample-to-Insight Workflow

6 stages · one validated pipeline



METAGENOMICS PORTFOLIO

ASSAYS (1-3) · SEQUENCING TECHNOLOGIES (4-5) · APPLICATION PANELS (6-9)

ASSAYS

1 16S rRNA Sequencing BACTERIAL PROFILING Cost-effective bacterial community profiling via conserved ribosomal markers. High-throughput diversity analysis across stool, soil, and water. Ideal for large-cohort microbiome studies. Archaeal profiling may be supported with appropriate primer selection.	2 ITS Sequencing FUNGAL COMMUNITY PROFILING Species-level fungal identification via ITS1/ITS2 amplicons. High-resolution mycobiome profiling across clinical, agricultural, and environmental matrices.	3 Whole Metagenome Shotgun WHOLE-COMMUNITY DNA Culture-free sequencing of all microbial DNA. Simultaneous taxonomy, AMR profiling, and functional pathway analysis for comprehensive community characterisation. - Multi-kingdom detection - Functional pathway analysis - AMR determinant profiling - MAG reconstruction
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SEQUENCING TECHNOLOGIES

4 Short-Read Sequencing GeneMind PLATFORM High-accuracy, high-depth reads for robust species ID and functional annotation. Scalable and cost-effective for large cohort studies. Compatible with 16S, ITS, and whole-metagenome strategies.	5 Long-Read Sequencing NANOPORE PLATFORM Kilobase-scale reads enabling complete genome assembly and strain-resolved taxonomy. Real-time pathogen ID with adaptive sampling. Full AMR gene chromosomal context. Compatible with 16S rRNA, ITS amplicon, and WMS workflows.
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APPLICATION PANELS

6 Clinical Metagenomics PATHOGEN DETECTION Culture-free multi-kingdom ID from blood, CSF, BAL, stool, and tissue. Concurrent AMR genotyping.	7 Gut & Urobiome Analysis INTESTINAL MICROBIOTA Comprehensive intestinal and urinary profiling linking dysbiosis to IBD, metabolic syndrome, and UTI. Supports therapeutic monitoring and restoration.
8 Plant & Rhizosphere Profiling RHIZOSPHERE & ENDOPHYTES Root-zone and internal plant microbiota profiling. Identifies growth-promoting bacteria, mycorrhizal fungi, and phytopathogens across rhizosphere and endophytes.	9 Environmental Surveillance SOIL · WATER · SEDIMENT Biodiversity assessment and AMR surveillance across terrestrial and aquatic matrices. Detects indicator organisms and novel pathways for regulatory compliance.

Advantages & Trade-Offs

what to weigh before you sequence

+ Advantages

- ▶ **Culture-independent** — Direct profiling from any sample.
- ▶ **Multi-kingdom** — Bacteria, fungi, viruses, archaea, protozoa.
- ▶ **AMR detection** — Resistance genes identified genome-wide.
- ▶ **Functional profiling** — Metabolic pathways and virulence factors mapped.
- ▶ **Strain-resolved** — Species-to-strain resolution with long reads.
- ▶ **Scalable** — Single samples to large population cohorts.
- ▶ **Broad matrix compatibility** — Clinical and environmental.
- ▶ **Fast turnaround** — Actionable results in 5-10 business days.

- Trade-Offs

- ▶ **Higher cost** — Shotgun exceeds amplicon-based pricing.
- ▶ **Host contamination** — Human reads dilute microbial signal.
- ▶ **Bioinformatics complexity** — Specialist expertise required.
- ▶ **Database limits** — Novel organisms may be missed.
- ▶ **Low biomass samples** — Sparse samples need enrichment.
- ▶ **Extraction bias** — Lysis protocols may skew community data.
- ▶ **Longer workflow** — Complex projects: 7-14 days report.
- ▶ **Expert interpretation** — Multi-kingdom data needs specialist review.

Sample Type Requirements

13 accepted matrices · colour coded by category

SAMPLE TYPE	APPLICATION	WHAT IT ENABLES
Stool	Gut microbiome · IBD · Oncology	Full community profiling; dysbiosis mapping; therapeutic monitoring
Blood	Sepsis · Bloodstream infections	Culture-free pathogen ID and AMR genotyping from primary material
Urine	UTI · Urobiome research	Low-biomass urobiome characterisation; fastidious uropathogen detection
Sputum	Pulmonary infections · Lung	Respiratory pathogen detection; lung dysbiosis in CF and COPD
Skin Swab	Dermatology · Wound infection	Skin microbiome diversity; wound pathogen ID and resistance genotyping
Tissue Biopsy	Tumour microbiome · Infection	Intratumoral microbial signatures; sterile-site infection source ID
Endometrial Tissue	Gynaecology · Fertility	Uterine microbiota; Lactobacillus dominance; infertility & endometriosis dysbiosis
Plant Tissue	Endophyte profiling · Phytobiome	Internal plant microbiota; beneficial vs. pathogenic colonisation
Rhizosphere Soil	Root-zone microbiome · Agriculture	Root-associated community; growth-promoting organism profiling
Soil	Ecosystem health · Soil fertility	Bulk biodiversity; carbon cycling genes; contamination assessment
Water	Drinking water safety · Ecology	Pathogen surveillance; indicator organisms; aquatic community diversity
by organism group · typical starting points		
Bacteria 5–10 M reads	Fungi 10–20 M reads	Archaea 5–10 M reads
Environmental 20–50 M reads	Shotgun MGX 20–50 M reads	Clinical MGX 30–100 M reads
Viruses 20–50 M reads		Protozoa 20–40 M reads
		Long-Read MGX 5–20 Gb data

Short-Read vs Long-Read Platform

choose based on your study goal

Short-Read · GeneMind	Long-Read · Nanopore
✓ Accuracy >99.9% per base (Q30+)	✓ Read length >1 kb–2 Mb · complete chromosomes
✓ Cost-effective at depth Lower cost per base	✓ Strain-resolved Full-length reads · high resolution
✓ Large cohorts Scalable, high-throughput	✓ AMR gene context Full chromosomal context
✗ Read length 150–250 bp · fragmented contigs	✓ Real-time output Adaptive sampling · 48 hrs
✗ Strain resolution Moderate · limited SVs	✗ Cost per base Higher than short-read
✗ AMR context Known alleles only	✗ Accuracy 95–99% per base (Q20+)
★ Best for Population studies · high depth · diversity	★ Best for Assembly · strain ID · AMR · novel orgs

Detection Capabilities

what a single run identifies

Bacteria	Fungi	Viruses	Archaea	Protozoa
AMR Genes	Virulence Factors	Mobile Elements	Functional Pathways	Novel Organisms

Values are general guidance, not fixed cut-offs. Optimal sequencing depth, QC thresholds, and input requirements shift with matrix complexity and study design. Our team tailors every workflow — extraction, library strategy, and bioinformatics pipeline — to your project before sequencing begins.